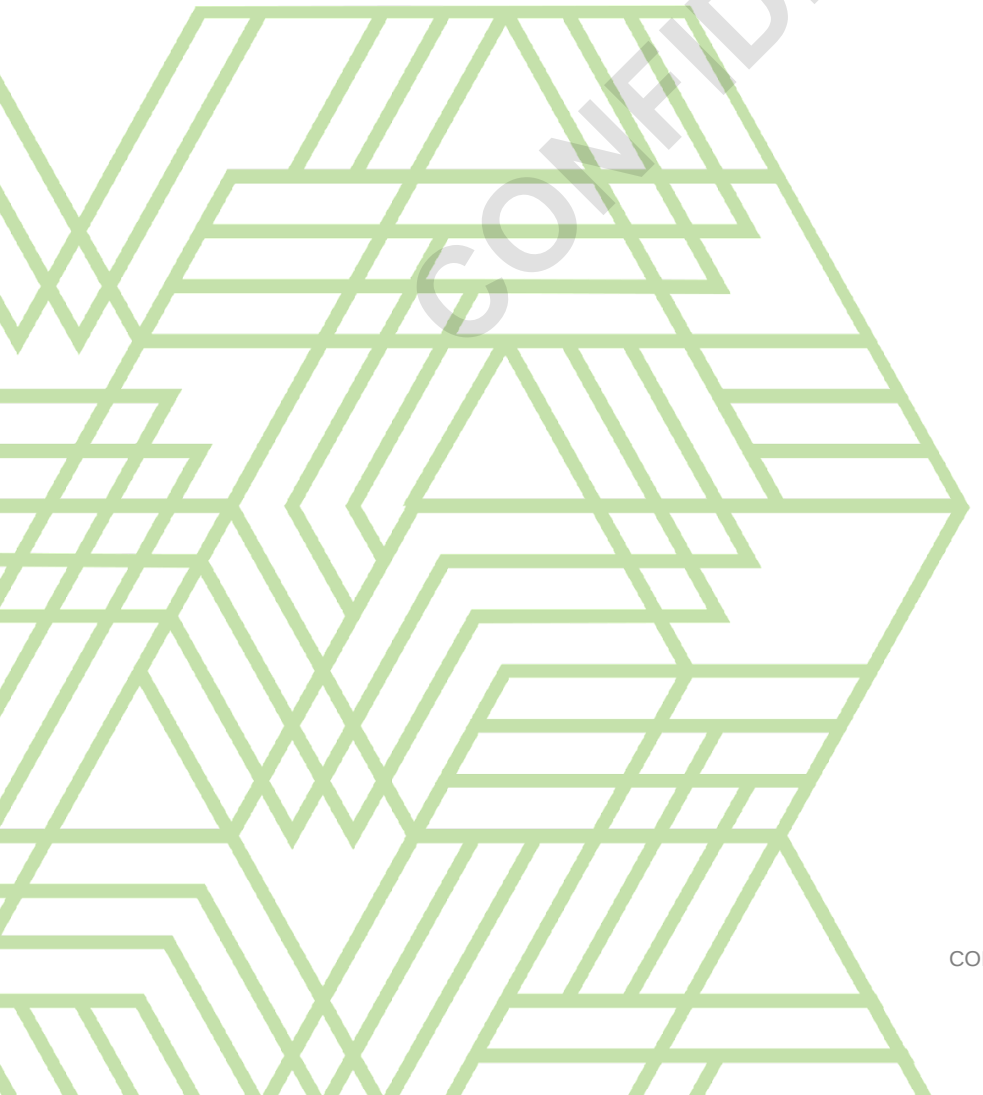
ATM33e Series
Energy Harvesting

Application Note

SUMMARY: This document describes the functionality of the ATM33e Wireless SoC Series harvester and performance of PV and RF harvesting. This document also includes a programming example to demonstrate how to enable harvesting settings using the Atmosic Software Development Kit (SDK).

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Table 1 - Applicable SoC and EVK

Acronyms and Abbreviations

Acronyms	Definition
ATM33e	ATM3330e
ATM33/e	ATM3325 ATM3330 ATM3330e
EVB	Evaluation Board
EVK	Evaluation Kit
PMU	Power Management Unit
PV	Photovoltaic
RF	Radio Frequency
SDK	Software Development Kit
TEG	Thermoelectric Generator

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1. Overview

The Atmosic ATM33e Wireless SoC series supports energy harvesting from RF sources or from other modalities such as PV, TEG, and mechanical actuation. Harvesting can be used with a battery (non-rechargeable or rechargeable) to extend its lifetime. For the specific harvesting modes and energy storage configurations supported, see the latest **Atmosic SDK Release Notes**.

This document describes the functionality of the ATM33e harvester and performance of PV and RF harvesting with a non-rechargeable battery. For the harvesting performance with lithium ion battery, please refer to **ATM33/e Support for Storage Devices Application Note**. This document also includes a programming example to demonstrate how to enable harvesting settings using the Atmosic SDK. Please refer to the **Atmosic SDK User Guide** for additional information on building and programming flash applications.

For information on EVB descriptions, please refer to **ATM33/e Series Evaluation Kit User Guide**.

Please refer to the Reference Documents section for related information.

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2. Hardware and Software Requirements

2.1 Supported EVK

EVK	SoC	SoC Part Number	Kit Part Number
Evaluation Kit for ATM3330e	56-pin 7x7 mm QFN	ATM3330E-5DCAQN	ATMEVK-3330e

Table 1 - Applicable SoC and EVK

***Note**: Energy harvesting feature is applicable to ATM33e Wireless SoC series only.*

2.2 Supported SDK

The Atmosic SDK 5.3.0 or later can be used with the EVK. Refer to the SDK User Guide for more information.

3. ATM33e Power Management and Harvesting

The ATM33e's integrated PMU is responsible for the energy transfer between the sources (harvesters and storage elements) and sinks (Bluetooth LE system and peripherals). It also generates the core and I/O supplies required for the SoC. There is a cold start energy requirement for the PMU to boot. After booting, the energy required to sustain operation depends on power consumption of the application.

Refer to **ATM33/e Support for External Storage Devices Application Note** for information on VBATLI pin as the input for a high-voltage battery or storage device.

3.1 PMU Configuration

The PMU must be configured properly to ensure correct operation. Please refer to the **ATM33/e Series Reference Manual**, **PMU Configurations** section for a list of supported PMU configurations. For their corresponding software settings, please refer to README in the SDK folder (platform/atm33/ATM33xx-5/examples/README). For software implementations, please refer to section Energy Harvesting Programming Example.

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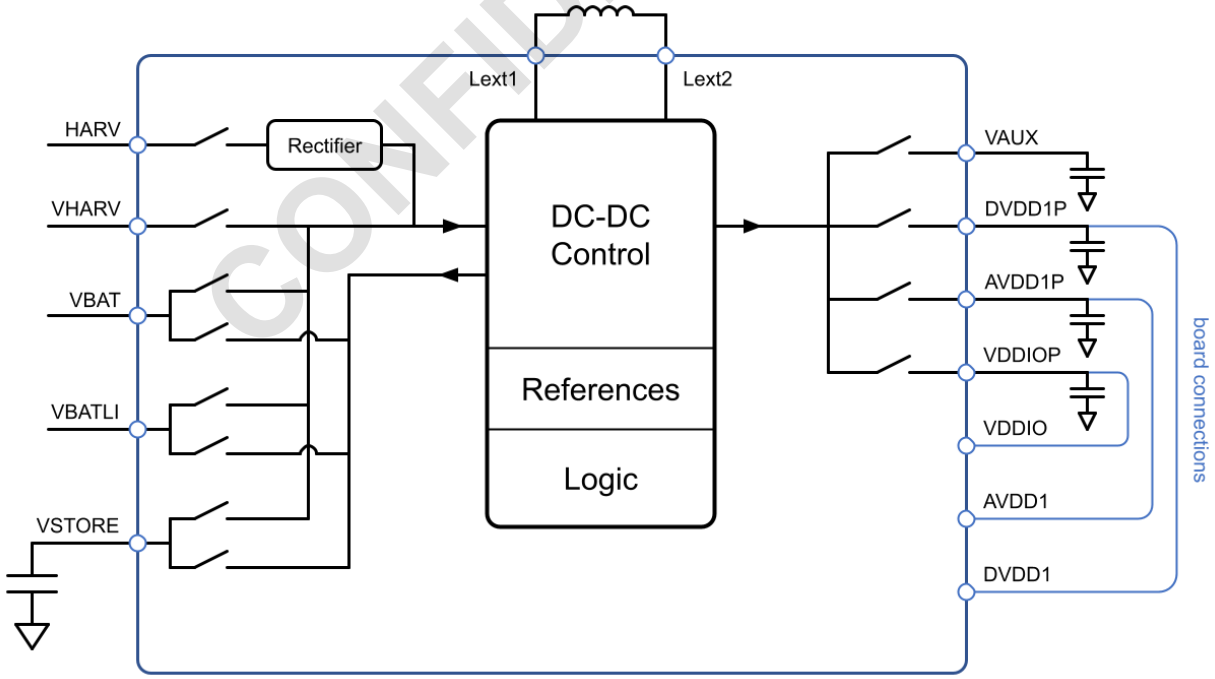


Figure 1 - Simplified Block Diagram of the PMU

3.2 Harvesting Description

3.2.1 Harvesting to load mode ON

This is the normal operation mode with harvesting. When there is incoming harvesting power, the ATM33e initially draws power from HARV (for RF harvesting) or VHARV (for non-RF harvesting, ie. PV harvesting). Excess power is transferred to the storage element connected to the VSTORE pin. The storage element can be a regular ceramic capacitor or a supercapacitor (the capacitive storage element connected to VSTORE will be referred to as Cstore for the rest of this document).

When harvesting power is not sufficient to meet the SoC load requirement, and there is enough energy on Cstore (i.e., VSTORE reaches VSTORE_START), power will be drawn from Cstore to support the load. When the Cstore energy reaches the upper limit (i.e., VSTORE reaches VSTORE_MAX), the harvester will shut off to prevent damage to the chip as well as the storage element.

In cases where the Cstore energy is not sufficient (i.e., VSTORE drops below VSTORE_MIN), the ATM33e will stop drawing power from VSTORE and start drawing from VBAT.

VSTORE_MAX is programmable (refer to section Energy Harvesting Programming Example). VSTORE_START and VSTORE_MIN are fixed at 1.7 V and 0.65 V respectively.

3.2.2 Harvesting to load mode OFF

This is a testing mode where the charging path and load path are separated. The incoming harvesting power is only used to charge Cstore. Cstore will continue to be charged until VSTORE reaches VSTORE_MAX.

ATM33e will draw power from VBAT, never from harvesting power directly or from VSTORE.

The charging efficiency in this mode will be discussed in the subsequent sections.

3.3 VSTORE Storage Element Considerations

The VSTORE storage element is an energy tank to ensure that constant power is available when needed for the application. It allows the system to handle any peak currents that cannot come directly from the harvesting source.

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A larger value of Cstore can sustain a larger power profile of the load with an intermittent harvesting source, but takes longer to reach VSTORE_START. It depends on the user application to select the optimal capacitor value.

4. ATM33e Harvester Performance

This section details the performance of the ATM33e for each of the supported energy harvesting types.

4.1 PV Harvesting

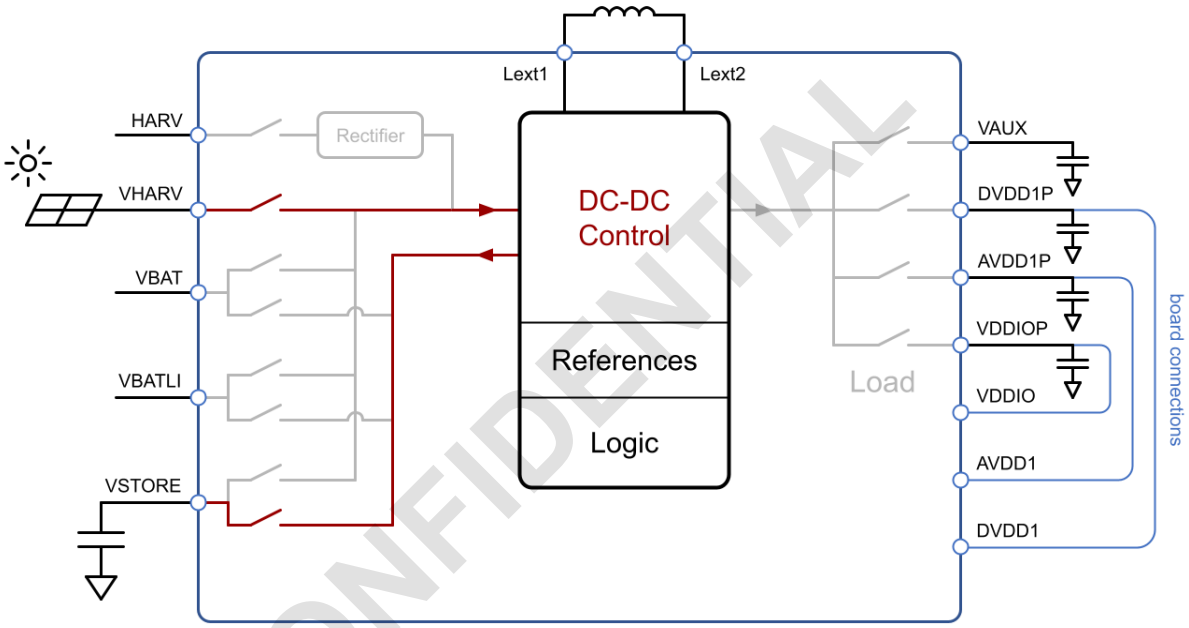


Figure 2 - PV Harvesting from VHARV to VSTORE (harvesting to load is Off)

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The PV cell operating voltage can be regulated by VHARV and can be programmed to one of the 16 settings from the lowest range of 0.4 V - 0.45 V to the highest range of 2.0 V - 2.5 V. Please refer to section Energy Harvesting Programming Examples for software implementations. For the full list of ranges, please refer to section Harvesting Type (ATM33e only) in the Reference Manual.

Ideal performance is achieved when using a PV cell that operates within the programmable range. The ATM33e cannot utilize power from the PV cell with an open-circuit voltage lower than the set minimum threshold (0.4 V - 0.45 V), and the ATM33e will not be able to utilize all the available energy from a PV cell with an operating voltage higher than the set maximum threshold (2.2 V - 2.5 V).

***Warning**: the PV harvested energy input port (VHARV) has a maximum input current rating. Exceeding the maximum input can potentially degrade the performance or permanently damage the device. Consult the ATM33e Series datasheet for more information.*

For the Panasonic PV cell (AM-1454) included in the ATM33e Evaluation Kit (datasheet), at indoor illumination levels (approximately 100 to 2000 lux), the voltage regulation range of 1.7 V - 2.0 V should be selected in the software, offering optimum power transfer from the PV cell to ATM33e.

Figure 3 shows the PV charging efficiency from VHARV input to VSTORE at various input light intensity levels with `harvesting to load` mode ON. This is the efficiency of the path shown in Figure 2. Efficiency here is defined as power to VSTORE vs power delivered by the PV cell. The voltage and current of the PV cell was monitored at every light intensity for this measurement. PV charging efficiency is continuously being improved through the upcoming software releases. Please contact Atmosic support on the information regarding latest improvements.

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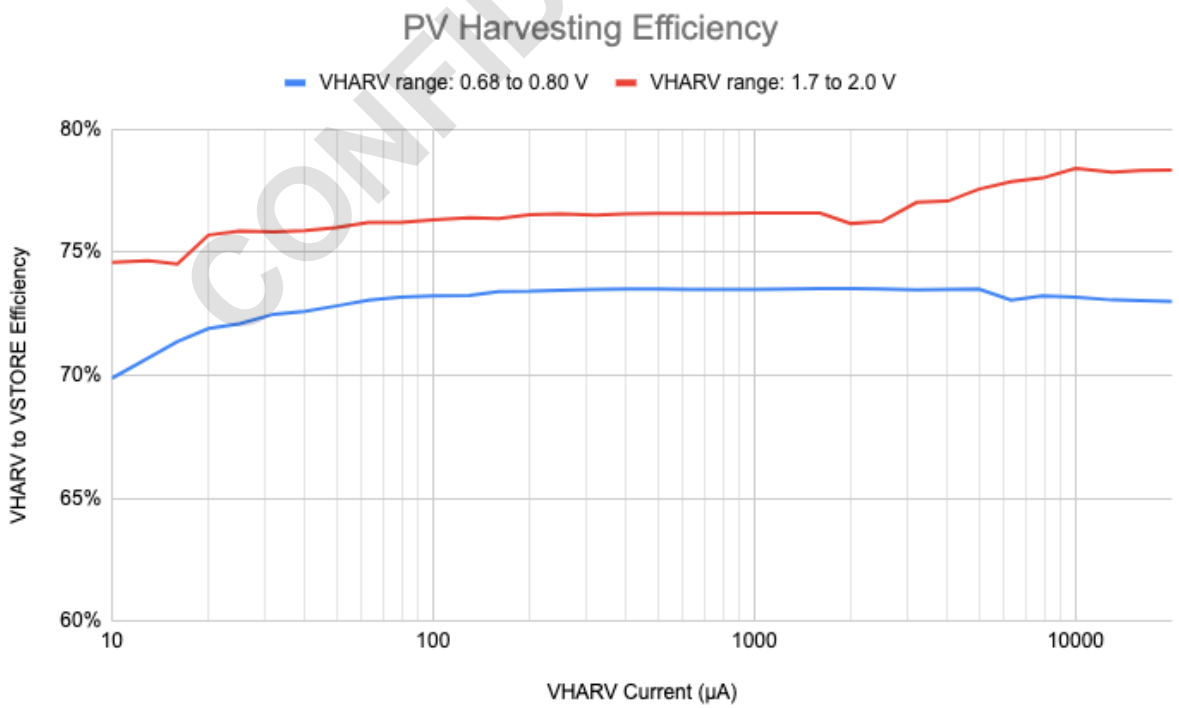


Figure 3 - Charging Efficiency from VHARV to VSTORE for Optimal VHARV Range

Figure 4 shows the power harvested under different indoor lighting conditions. The power harvested is the power that is sent to VSTORE.

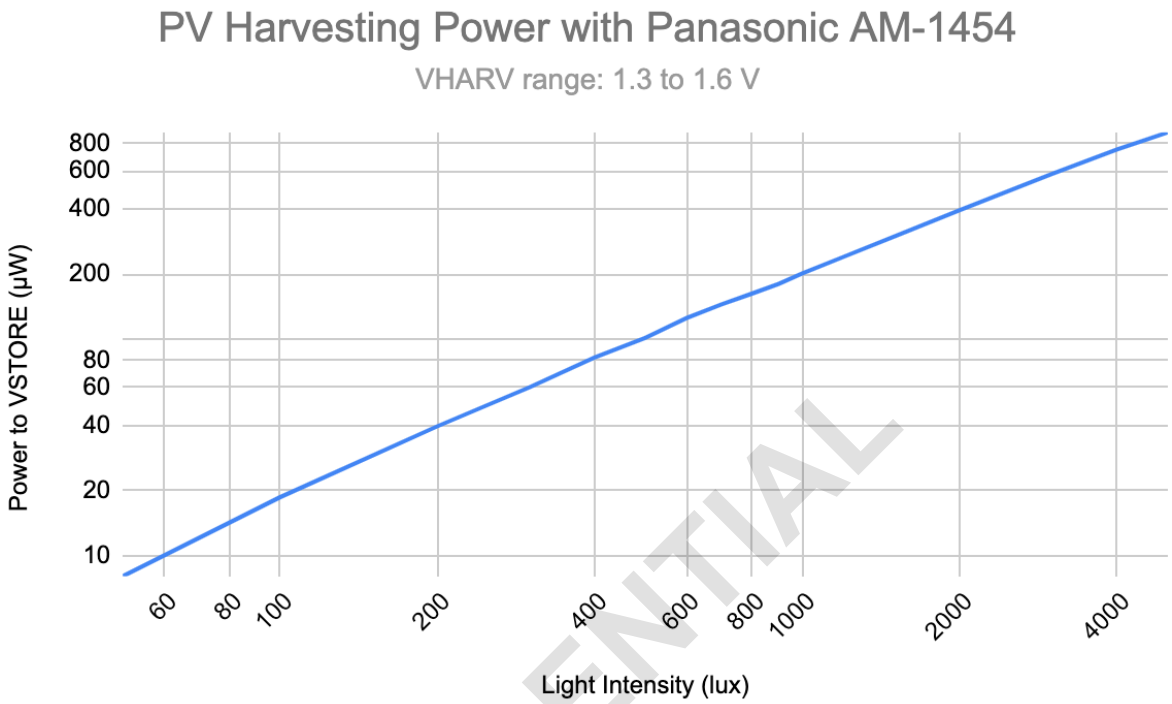


Figure 4 - Power Harvested for Different Lighting Intensity.

With `harvesting to load` mode ON, under indoor lighting, the user can observe that VBAT current consumption is dropping, indicating the load current is supplied by the harvesting source and Cstore.

4.2 RF Harvesting

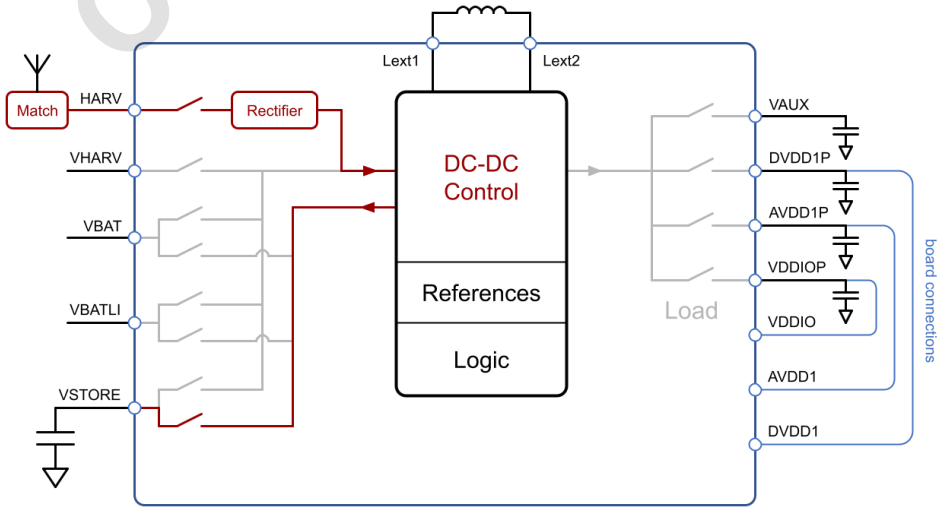


Figure 5 - RF Harvesting from HARV to VSTORE (`harvesting to load` is off)

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Warning: The RF input port (HARV_INN and HARV_INP, shown as HARV in Figure 5) has a maximum rating of +10 dBm. Exposure to a higher input RF power can potentially degrade the chip performance and damage the chip.

Figure 6 shows the RF charging efficiency from the HARV SMA connector (HARV_INN and HARV_INP, referred to as HARV in Figure 6) to VSTORE at various HARV input power levels with `harvesting to load` mode ON. The data is acquired on two ATM33e Evaluation Boards with 2.4 GHz match and 915 MHz match respectively. The measured efficiency in Figure 6 is the efficiency of the path shown in Figure 5. It is the efficiency from HARV SMA to VSTORE, which includes the matching network (on EVB and not port of the ATM33e), rectifier, and the DC to DC converter.

The RF harvester design supports Maximum Power Point Tracking (MPPT), which automatically sets the optimal settings as RF power changes.

For different board designs and different RF source frequencies, it is required to customize the on-board matching components for the optimal harvesting performance. Please submit a support request to Atmosic for more information.

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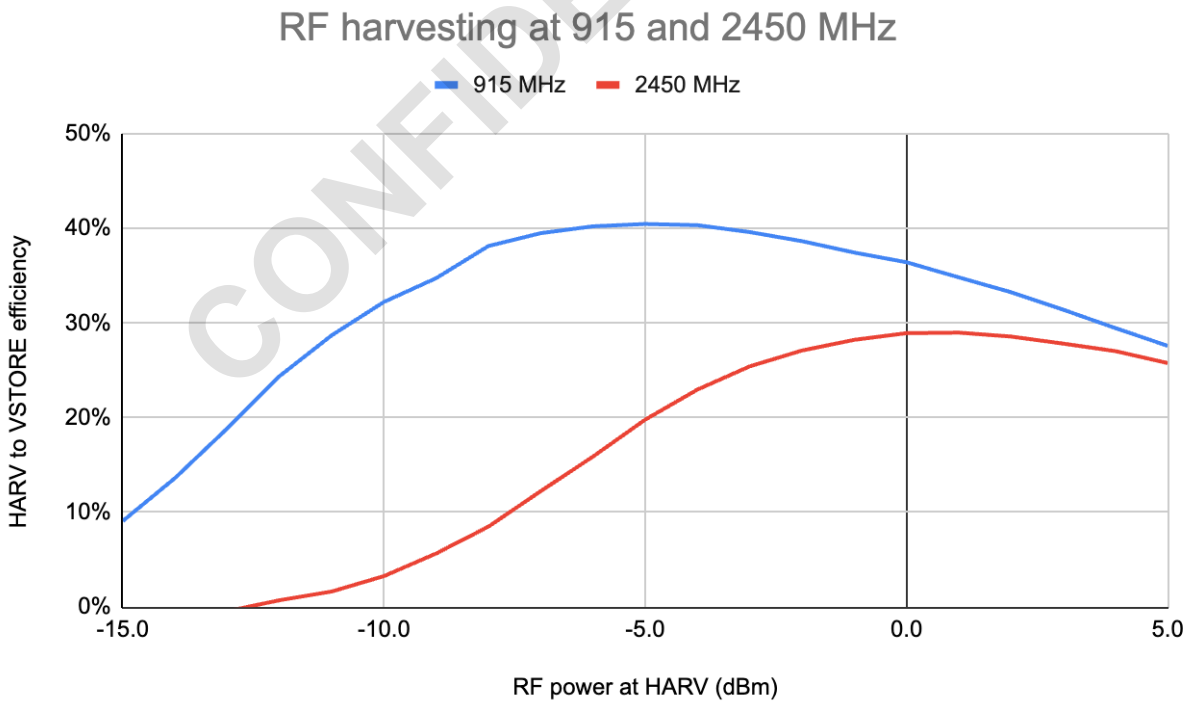
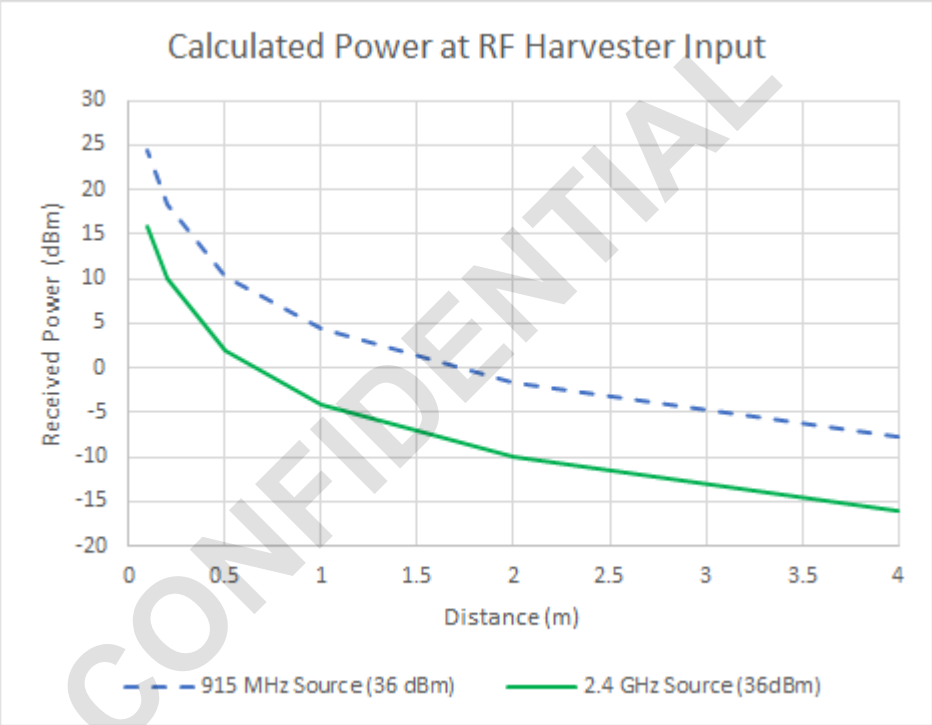


Figure 6 - Charging efficiency from HARV SMA to VSTORE

With `harvesting to load` mode ON, and RF input power available, the user can observe that VBAT current consumption is dropping, indicating the load current is supplied by the harvesting source and Cstore.

As part of the system design, a RF harvesting source should be considered carefully to support the application load. The received power at the antenna can be estimated by the free-space path loss, as demonstrated in Figure 7. From a 36 dBm (4 Watt) source, Figure 7 below shows the calculated received power versus distance for both 915 MHz and 2.4 GHz frequencies.



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Figure 7- Power at RF Harvester Input

4.3 Combined Harvesting

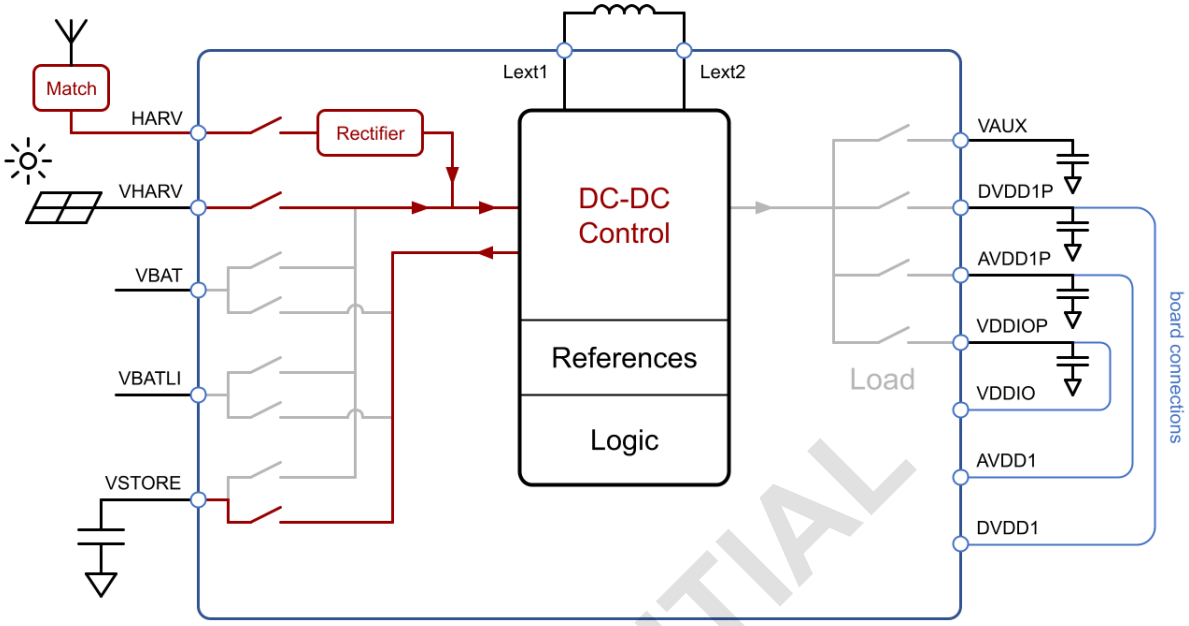


Figure 8 - Combined Harvesting from VHARV and HARV to VSTORE (`harvesting to load` is Off)

In combined harvesting, both RF and PV energy could be harvested simultaneously. Similar to PV only harvesting, the PV cell operating voltage can be regulated by VHARV, and can be programmed to one of the 16 settings. No extra settings are needed for RF harvesting except for activating RF harvesting in the SDK. Everything said in the two sections above on RF harvesting and PV harvesting applies to combined harvesting. RF harvesting efficiency will be different when using combined harvesting than when using RF only harvesting. In combined harvesting, the RF harvesting efficiency depends on the voltage at the VHARV pin which is set by VHARV_RANGE and is lower than RF only harvesting as it uses MPPT.

5. Energy Harvesting Meter

ATM33e has a built-in harvesting meter, which would tell the user the amount of energy being harvested. The harvesting meter works with PV and combined harvesting, but not with RF only harvesting.

The PMU is a switching regulator and it moves energy around (from a source to a destination) in small chunks repeatedly as the energy is needed at a destination or in the case of harvesting when the energy is available at the harvester. The harvesting meter

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monitors the PMU and, in a fixed amount of time, counts the number of times the PMU moves energy from the harvester (VHARV), the source, to the storage (VSTORE), the destination. This timed count is directly proportional to the total harvested energy since each transfer is a fixed and known amount of energy. And finally, by dividing the total energy by the time used for counting, we get the power being harvested.

The harvesting meter does the counting/measurement right after the ATM33e goes into a sleep state (hibernation or retention). When the chip comes out of a sleep state and back into active state, the measured harvesting power will be available to be read by the user.

It should be noted that the harvesting meter has some limitations. It only measures the harvesting power for the first few hundred milliseconds immediately after going into a sleep state. If the chip stays in a sleep state for long and the harvested energy changes significantly during that time, it will not be captured by the harvesting meter. The more often the chip goes in and out of a sleep state, the more accurate the overall average will be. In most cases with PV harvesting, the light condition in the room doesn't change significantly within seconds, or even minutes, and so the results would be very close to the actual harvested power.

The harvesting meter has one input parameter which is the value of the switching regulator inductor, which is connected to LEXT1 and LEXT2 pins. The value is in µH. The reason the software needs to know this value is because the amount of each energy transfer is inversely proportional to the value of the external inductor.

The harvesting meter is inactive by default. The user has to activate it using software. When using the examples in the SDK, the user can activate the harvesting meter by setting the CFG_HARV_METER flag to one. Of course, this only works if either PV or combined harvesting has also been activated by software. Section 6 covers examples of how to activate harvesting using the SDK examples. Each time the chip comes out of a sleep state, the software will report both the instantaneous power measured in the last sleep state, and the average of all harvested power measured. If the harvesting meter is disabled and enabled again, the average harvested power is reset. The power reported is in µW.

The harvesting meter can only work when energy is being transferred to the energy storage at VSTORE. If the storage is full, the excess energy is dumped to ground, and the harvesting meter can't do the counting. In a case where the storage is full, the measurement is invalid and the software will return invalid measurement and it will not be included in the average. Although this shouldn't be a problem for most users since if

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the storage is full, we often don't need or care about harvesting until some or most of the energy in the storage has been used.

In lab testings, the harvesting meter was accurate to within 10% of the actual harvested power. At low harvesting powers (100 lux of light or less with a typical PV cell such as the Panasonic AM-1454), the accuracy drops to less than 10%.

6. Energy Harvesting Programming Examples

The user can enable energy harvesting by indicating battery options and harvesting mechanism in the makefile of one of the SDK examples. For all the settings supported by the SDK, please refer to the README in the SDK specific applications folder (platform/atm33/ATM33xx-5/examples/).

A programming example is included below to demonstrate how to modify the makefile for energy harvesting evaluations. For instructions on how to install the SDK and program the EVB, please refer to the **Atmosic SDK User Guide** (available on the Atmosic Support website).

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6.1 PV Harvesting Example

The easiest way to test harvesting is to enable harvesting in an SDK example and observing or measuring the harvested energy between beacons. Different examples in the SDK, ATM33e can send periodic scannable beacons. The time between each beacon is set by the beacon profile in the `reference_beacons.mk` file, but it could also be set with the makefile command. In the `BLE_adv` example, the default is `simple_beacon`, which transmits a beacon every 100 ms. To test harvesting, it is more convenient to use the `simple_harv_beacon`, which transmits a beacon every second.

The example below is for PV harvesting. It assumes the PV cell performs best between 1.7 to 2.0 V. And that VBAT is connected to a 3 V supply. If different settings are needed, the user can adjust the flags accordingly. For other options, please refer to the **Atmosic SDK User Guide**.

The corresponding compilation flags (CFLAGS) are shown below. The makefile flags enable PV harvesting energy (Figure 9 blue) with a battery (Figure 9 green) in the system. VSTORE is set to regulate at 3.3V as it is the maximum allowed level (Figure 9 white). The PV cell is configured for a desired operating voltage from 1.3 to 1.6 (Figure 9 red).

Steps:

1) In the SDK example `BLE_adv` folder (platform/atm33/ATM33xx-5/examples/BLE_adv/), modify and insert the additional flags into the CFLAGS section of the makefile as shown in Figure 9.

2) Run `make clean`

3) Run `make BOARD=ATMEVK_3330e_QN REF_BCN:=simple_harv_beacon run_all` to build the application and load it to the target.

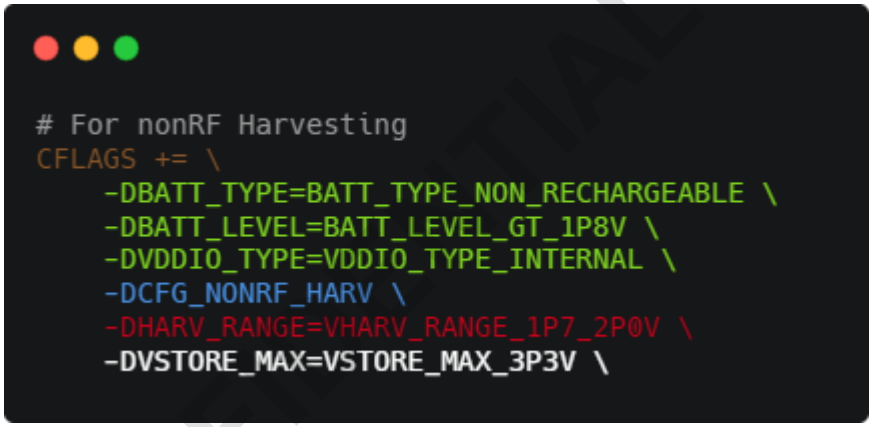


Figure 9 - Examples/BLE_adv/makefile for PV Harvesting

Reference Documents

Title	Document Number
ATM33e Series Datasheet	ATM33e-DS
ATM33e Series Energy Harvesting EVK Quick Start Guide	ATM33e-QSGEHV
ATM33/e Series Reference Manual	ATM33_e-RM
ATM33/e Series Evaluation Kit User Guide	ATM33_e-UGEVK
ATM33/e Support for External Storage Devices Application Note	ATM33_e-ANSESD
Atmosic SDK Release Notes	ATM-RNSDK
Atmosic SDK User Guide	ATM-UGSDK

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Revision History

Date	Version	Description
March 10, 2023	0.61	Added Combined Harvesting, Figure 4 - Power Harvested for Different Lighting Intensity, Energy Harvesting Meter. Updated Overview, Harvesting to load mode ON, PV Harvesting, RF Harvesting, Combined Harvesting, Energy Harvesting Programming Example, PV Harvesting Example sections, Figure 3 - Charging Efficiency from VHARV to VSTORE for Different VHARV Ranges, Figure 6 - Charging efficiency from HARV SMA to VSTORE, Figure 9 - Examples/BLE_adv/makefile for PV Harvesting.
August 17, 2022	0.60	Updated Related Documents, PMU Configuration, PV Harvesting, RF Harvesting, and Energy Harvesting Programming Example sections.
December 3, 2021	0.51	Changed part number, no content change..
November 9, 2021	0.50	Initial version created.

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